

Optimizing Portfolio with Two-Sided Transactions and Lending: A Reinforcement Learning Framework

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Abstract

This study presents a Reinforcement Learning (RL)-based portfolio management model tailored for high-risk environments, addressing the limitations of traditional RL models and exploiting market opportunities through two-sided transactions and lending. Our approach integrates a new environmental formulation with a Profit and Loss (PnL)-based reward function, enhancing the RL agent's ability in downside risk management and capital optimization. We implemented the model using the Soft Actor-Critic (SAC) agent with a Convolutional Neural Network with Multi-Head Attention (CNN-MHA). This setup effectively manages a diversified 12-crypto asset portfolio in the Binance perpetual futures market, leveraging USDT for both granting and receiving loans and rebalancing every 4 hours, utilizing market data from the preceding 48 hours. Tested over two 16-month periods of varying market volatility, the model significantly outperformed benchmarks, particularly in high-volatility scenarios, achieving higher return-to-risk ratios and demonstrating robust profitability. These results confirm the model's effectiveness in leveraging market dynamics and managing risks in volatile environments like the cryptocurrency market.

Keywords: Portfolio Management; Reinforcement Learning, Deep Learning, Context-driven Optimization; Cryptocurrency Market

JEL classification: C45; C61; G11; G17

1. Introduction

The escalating complexity of today's financial markets, characterized by the globalization of financial activities, the rapid growth in the number and types of financial products (such as derivatives and complex investment vehicles), the increased use of technology and algorithmic trading, and the interconnectedness of global economies and financial systems, has challenged the adequacy of traditional portfolio management models. They struggle with adapting to new market features and complexities, relying on single-period or oversimplified multi-period modeling. This shortfall is particularly noticeable in highly non-stationary markets. [Li et al. \(2023a\)](#).

To address these challenges, machine learning (ML) approaches have been increasingly applied to portfolio optimization e.g. [Mulvey et al. \(2020\)](#), [Barua and Sharma \(2022\)](#), [Lin and Taamouti \(2023\)](#), [Liu et al. \(2023b\)](#) and [Barua and Sharma \(2023\)](#). However, these ML models often depend on predicting price trends, which, as shown in various studies like [Moody and Saffell \(2001\)](#), may not necessarily correlate with maximizing portfolio performance. This realization has sparked interest in exploring model-free approaches, such as Reinforcement Learning (RL), known for their efficacy in policy-based problems across diverse domains [Cui et al. \(2023\)](#). RL-based portfolio management models uniquely integrate prediction and weight management into a unified process and are capable of optimizing portfolio performance for varied objectives and constraints.

Expanding on the definition of RL by [Sutton and Barto \(2018\)](#), a key aspect of portfolio management within this framework lies in the understanding of the environment states, which is analogous to the historical data of assets. Decisions or actions in this context involve transactions to rebalance the assets' weights in the portfolio. The crux of these actions is underpinned by the reward, defined as scalar functions of the financial earnings or losses resulting from these transactions.

There have been significant advances in applying RL to the portfolio management problem in recent years, but the field has not fully explored the effective use of new market facilities in the model. Integrating advanced market features like two-sided transactions and lending into RL adds considerable risk and complexity, resulting in the response's non-convergence or undesirable outcomes. Our study aims to tackle this issue by developing a new environmental formulation with a Profit and Loss (PnL)-based reward function. By focusing on key constraints like transaction costs, and employing the Soft Actor-Critic (SAC) agent with a Convolutional Neural Network with Multi-Head Attention (CNN-MHA) architecture, we enhance the model's ability to downside risk management and capital optimization.

The SAC agent ([Haarnoja et al. \(2018\)](#)), a model-free, actor-critic algorithm, aims to maximize both reward and entropy, with entropy serving as a gauge for policy uncertainty. This dual focus fosters an equilibrium between reward maximization and environmental exploration, granting the agent the resilience to effectively

adapt to novel market scenarios. The MHA architecture (Vaswani et al. (2017)) complements the SAC agent by transcending the constraints of traditional Recurrent Neural Networks (RNNs). The MHA architecture relies on the self-attention mechanism, which enables comprehensive contextual processing of sequential data Li et al. (2023b). This mechanism allows each element in a sequence to relate to every other element, effectively overcoming the limitations of dependency in previous sequence networks like Long Short-Term Memory (LSTM).

In the implementation phase, we utilize a diversified 12-crypto asset portfolio in the Binance perpetual futures market as the RL environment to train and test our model. The cryptocurrency market, with unique characteristics of continuous 24-hour transactions, decentralization, low transaction fees, high security through blockchain technology Fang et al. (2022), and significant tail-risk exposure Borri (2019), presents a challenging but appropriate landscape for implementing and evaluating ML-based portfolio management models Ren et al. (2022).

The remainder of this paper is organized as follows: Section 2 explores existing portfolio management models in literature. Section 3 details the research methods and data used. Section 4 presents the empirical results, including an analysis of test results compared to benchmark models. The paper concludes with a summary of the study's findings in Section 5.

2. Literature Review

In recent years, building on Markowitz (1952)'s research, numerous studies have focused on mathematical modeling to tackle the portfolio management problem. These efforts can be divided into two main categories: Single-Period Portfolio Optimization (SPPO) and Multi-Period Portfolio Optimization (MPPO) models.

SPPO models adopt a static approach, optimizing portfolio weights at the outset of the investment period without further modifications. A significant evolution in this category is the introduction of alternative risk measures in place of variance, as proposed in the original Markowitz framework. Innovations include Variance with Skewness (VwS) by Samuelson (1958), Semi-Variance (S-V) by Markovitz (1959), Mean-Absolute Deviation (MAD) by Konno and Yamazaki (1991), Value-at-Risk (VaR) by Jorion (1997), Minimax (MM) by Young (1998), and Conditional Value-at-Risk (CVaR) by Rockafellar et al. (2000). These models typically employ quadratic or linear programming for Efficient Frontier (EF) determination, with several studies have explored multi-objective optimization methods to discover a more optimal EF, e.g., Roman et al. (2007), Tsao (2010), Petchrompo et al. (2022) and Wang et al. (2023). However, SPPO models are often critiqued for their reliance on historical data and oversimplified assumptions (Michaud (1989)), limiting their applicability in multi-stage and long-term investment scenarios.

Conversely, MPPO models, adhering to Kelly (1956)'s optimal capital growth approach, allow periodic reassessment and readjustment of portfolio weights. This

category, which began with the contributions of [Smith \(1967\)](#), [Mossin \(1968\)](#), [Merton \(1969\)](#), [Chen et al. \(1971\)](#) and [Fama \(1975\)](#) researches and was further advanced by [Li and Ng \(2000\)](#) and [Zhou and Li \(2000\)](#), explored in both continuous and discrete time frameworks, leverages Dynamic Programming (DP), Stochastic Programming (SP), and Stochastic Dynamic Programming (SDP) to find optimal portfolio weights.

DP-based MPPO models, ideal for long-term investments with frequent rebalancing, face computational complexity and reliance on simplifications due to practical constraints like transaction costs, e.g., [Kraft and Steffensen \(2013\)](#), [Cui et al. \(2017\)](#) and [Das et al. \(2022\)](#). On the other hand, SP-based models, incorporating practical constraints, offer a more realistic approach but face computational challenges in long-term scenarios, e.g., [Gaivoronski and Stella \(2003\)](#), [Fulton and Bastian \(2019\)](#) and [Barro et al. \(2022\)](#). SDP-based models, including RL, integrate the benefits of DP and SP approaches, making them suitable for long-term investment strategies involving random variables, e.g., [Topaloglou et al. \(2008\)](#), [Luo and Zeng \(2014\)](#), and [Van Staden et al. \(2018\)](#).

The application of RL in portfolio management models first gained attention through [Neuneier \(1997\)](#), which utilized the Q-Learning agent to identify optimal portfolio management policy. Following this, [Neuneier and Mihatsch \(1998\)](#) enhanced the Q-Learning by incorporating risk assessment. Later on [Gao and Chan \(2000\)](#) developed a Q-Learning-based portfolio management model aimed at maximizing the Sharpe ratio. Further advancements were made by [Moody and Saffell \(2001\)](#), who introduced Recurrent Reinforcement Learning (RRL) for portfolio management, utilizing the Direct Reinforcement (DR) algorithm.

Recent contributions in this area can be divided into the five key elements of an RL-based model: the Agent, the Environment, Actions, States, and the Reward function, focused on improving weight optimization.

The agent, comprises two essential components: the policy and the learning algorithm. The policy, often implemented through Neural Networks (NN), determines portfolio weights based on the current state. Notable recent research in NN design includes articles [Li et al. \(2023a\)](#), and [Zhao et al. \(2023\)](#), with a focus on the use of attention mechanism. The learning algorithm updates the policy using a trajectory of states, actions, and rewards. Popular learning algorithms in portfolio management, as outlined in [Hambly et al. \(2023\)](#), include Q-learning, DQN, DDQN, DPG, DDPG, and A2C, with several articles like [Halperin \(2019\)](#), [Jaisson \(2022\)](#), [Abousalah et al. \(2022\)](#), [Coache and Jaimungal \(2023\)](#), and [Mounjid and Lehalle \(2023\)](#) offering customized learning algorithms for financial contexts.

The environment integrates assets and market characteristics with a defined strategy for portfolio optimization. Notable recent developments in this field include [Wang and Zhou \(2020\)](#)'s implementation of the mean-variance model in a continuous-time format using RL, [Wu et al. \(2021\)](#)'s formulation of an environment based on the Equity Market Neutral (EMN) strategy, [Jin \(2023\)](#)'s implementation

of the mean-VaR strategy with two-sided transactions using RL, [Liu et al. \(2023a\)](#)'s employment of RL to discover nonlinear relationships and refine pairs trading strategies, and [Cui et al. \(2023\)](#)'s efforts to reduce the tail risk of digital currency portfolios by integrating the mean-CVaR strategy with RL.

Actions or optimal portfolio weights are divided into discrete (buy, hold, sell) and continuous (direct weight determination) approaches, based on the characteristics of the agent. Discrete actions, though limited in precision, often incorporate complex reward functions with multiple constraints to address this limitation, as utilized in [Jaisson \(2022\)](#), and [Ngo et al. \(2023\)](#). Conversely, continuous actions improve accuracy by allowing the model to directly optimize portfolio weights. However, the majority of research using this approach does not account for negative weights and tends to concentrate on the spot market, like [Betancourt and Chen \(2021\)](#), and [Li et al. \(2023a\)](#) works.

states often refer to historical asset data. In the financial ML context, data preprocessing aimed at enhancing model performance involves signal decomposition techniques like Discrete Wavelet Transform (DWT) and Empirical Mode Decomposition (EMD), as shown in articles [Cheng and Wei \(2014\)](#), [Wang et al. \(2018\)](#), [Faria and Verona \(2018\)](#), [Lee et al. \(2021\)](#), [Xu and Wang \(2023\)](#), [Tang et al. \(2024\)](#), [Stein \(2024\)](#) and the application of technical indicators, similar to [Gradojevic and Gençay \(2013\)](#), [Liu et al. \(2022\)](#), [Ma and Yan \(2022\)](#), [Jang and Seong \(2023\)](#), [Tan and Tao \(2023\)](#) and [Goutte et al. \(2023\)](#), in addition to data normalization.

The *reward function* serves as the objective in optimizing portfolio weights. Portfolio return, Sharpe ratio, and PnL, as noted in [Hambly et al. \(2023\)](#), are commonly used reward functions in the literature. Additionally, several studies, such as [Halperin \(2019\)](#), [Jaisson \(2022\)](#), and [Aboussalah et al. \(2022\)](#), have developed specialized reward functions to optimize specific goals.

The literature review reveals a gap in research on negative weights and lending in RL-based portfolio management models, attributed to the increased risk these features introduce. This gap suggests a need for redesigning the environment and reward functions to enable the model to manage risk and capital more effectively. However, most existing studies opt for standard environments and conventional reward functions, focusing on optimizing networks and learning algorithms to improve performance.

3. Methodology

3.1. Reinforcement Learning

RL can be conceptualized as a tuple (S, A, R_a, P_a, γ) , comprising:

- S : The set of state representations.
- A : The set of actions available to the agent.
- R_a : The immediate reward received after transitioning to a new state due to action a .
- P_a : The state transition probability, expressed as $P_a(s', s) = P_r(s_{t+1} = s' \mid s_t = s, a_t = a)$, indicating the likelihood of transitioning to state s' from state s due to action a at time t .
- γ : The discount factor (0 to 1) signifies the importance of past actions.

The agent aims to find an optimal policy mapping states to actions to maximize cumulative discounted rewards, as defined in Cui et al. (2023). Figure 1 illustrates the components of the RL-based portfolio management model.

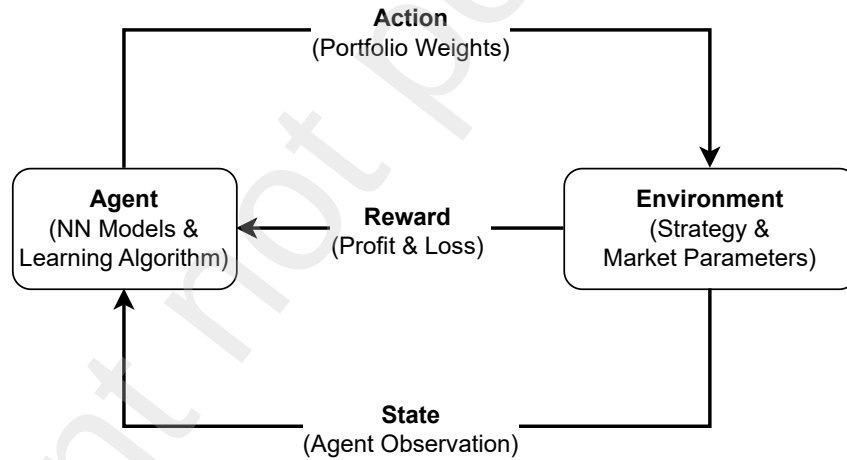


Figure 1: Architecture of reinforcement learning in portfolio management.

3.2. Data Preprocessing

Given the non-stationarity and varied scales of asset prices, preprocessing is necessary before network application. This process (Equation 1) involves normalizing and converting price data into a tensor (T) , with $M \times 4 \times N$ dimension in each state. Here, M represents the number of portfolio assets, 4 the number of features (open, high, low, and close prices), and N the historical data length for each feature.

$$\begin{aligned}
T_t &= [P_t^1, P_t^2, \dots, P_t^{M-1}, P_t^M] \\
P_t^i &= [Open_t^i, High_t^i, Low_t^i, Close_t^i] \\
Open_t^i &= \left[0, \frac{open_{t-n+2}^i}{open_{t-n+1}^i} - 1, \dots, \frac{open_{t-1}^i}{open_{t-n+1}^i} - 1, \frac{open_t^i}{open_{t-n+1}^i} - 1 \right] + noise \\
High_t^i &= \left[0, \frac{high_{t-n+2}^i}{high_{t-n+1}^i} - 1, \dots, \frac{high_{t-1}^i}{high_{t-n+1}^i} - 1, \frac{high_t^i}{high_{t-n+1}^i} - 1 \right] + noise \\
Low_t^i &= \left[0, \frac{low_{t-n+2}^i}{low_{t-n+1}^i} - 1, \dots, \frac{low_{t-1}^i}{low_{t-n+1}^i} - 1, \frac{low_t^i}{low_{t-n+1}^i} - 1 \right] + noise \\
Close_t^i &= \left[0, \frac{close_{t-n+2}^i}{close_{t-n+1}^i} - 1, \dots, \frac{close_{t-1}^i}{close_{t-n+1}^i} - 1, \frac{close_t^i}{close_{t-n+1}^i} - 1 \right] + noise
\end{aligned} \tag{1}$$

T_t comprises M price vectors (P_t) for each asset at period t , with P_t^i representing the price vector for asset i at period t , including four dimensions, $Open_t^i$, $High_t^i$, Low_t^i , and $Close_t^i$. These dimensions are normalized to reflect relative price changes, with additional scaled normal noise introduced to make the agent more resilient to market noise.

3.3. RL Agent

We employ the SAC agent, which utilizes two networks called the actor and the critic. the actor, determining the optimal weight rebalancing policy, while the critic guides the actor by evaluating environment rewards to maximize long-term discounted rewards and entropy. The actor's final layers output the mean and standard deviation of the action's normal distribution, from which the optimal weights are randomly selected. However, for model testing, we select actions with the highest likelihood as the portfolio weights.

In the development of the deep NN architectures for an M asset portfolio, as illustrated in Figure 2, we enhanced the input data by incorporating two CNN layers equipped with 64 and 128 filters, respectively, for feature extraction. Additionally, integrating an MHA layer, which highlights the more important parts of the sequence through an attention mechanism, boosts the agent's efficiency. In addition, the designed architectures utilize Fully Connected (FC) layers, incorporate the LeakyReLU activation function across all intermediate layers, and apply a unique optimized Learning Rate Factor (LRF) for each layer.

3.4. RL Environment

The RL environment primarily undertakes two key tasks: calculating the reward associated with each action taken and transitioning the agent to the next state through the defined strategy.

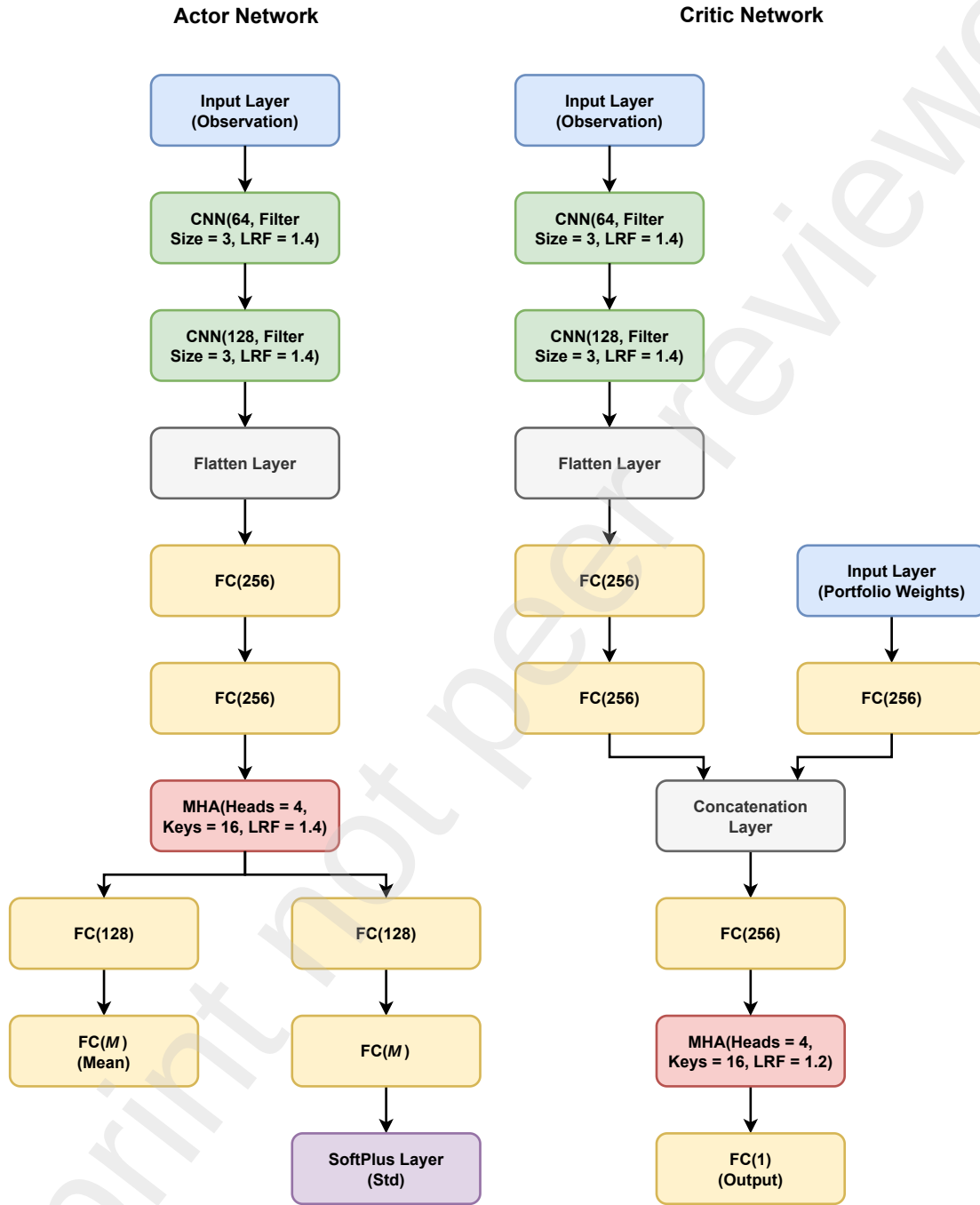


Figure 2: The architecture of the actor and critic networks.

At each state, the agent rebalances the portfolio weights using a vector W_t with $1 \times (M + 1)$ dimension. In this vector, $\forall w_t^i \in W_t, w_t^i \in [-1, 1], \sum_{i=1}^m |w_t^i| = 1$, and w_t^{m+1} signifies the loan amount. A positive value of w_t^{m+1} indicates the allocation of a portion of cash as a loan to accrue interest, while a negative value implies taking out a loan, increasing cash in the portfolio, and incurring interest.

When opening or closing positions in exchanges and brokers, a transaction fee (δ) is deducted. However, typically, there are no fees associated with either granting or receiving loans, and only the interest rate (ζ) applies. With these explanations, portfolio rebalancing calculations are presented by Equations 2 to 7.

For each state, the available rebalancing capital (c_t) is determined by multiplying w_t^{m+1} by the portfolio value at time t (p_t).

$$c_t = p_t \times (1 - w_t^{m+1}) \quad (2)$$

Next, the interest incurred from either granting (positive value) or receiving (negative value) a loan over the rebalancing interval is computed.

$$interest_t = p_t \times w_t^{m+1} \times \zeta \quad (3)$$

Given the current asset values in the portfolio ($Port_{t-1}$), the necessary changes according to W_t , and the related transaction costs ($Cost_t$), the readjusted asset values ($Port'_t$) are calculated.

$$Port'_t = c_t \times W_t^{1:m} - Cost_t \quad (4)$$

$$Cost_t = \delta \times (c_t \times W_t^{1:m} - Port_{t-1}) \quad (5)$$

Finally, the $Port_t$ value in the next state is calculated by multiplying $Port'_t$ by the assets' return (R_t),

$$Port_t = Port'_t \times (1 + R_t) \quad (6)$$

and the portfolio value at time $t + 1$ is obtained as p_{t+1} .

$$p_{t+1} = p_t + \sum (Port'_t \times R_t) + interest_t \quad (7)$$

The environment also computes the reward associated with the agent's action to transition to the next state. The reward function, as outlined in Equation 8, is designed to update the agent's policy to maximize rebalancing profit while controlling losses during training. Furthermore, to prevent the agent from making excessive modifications to the portfolio weights, transaction costs are incorporated into the reward function.

$$reward_t = \frac{\sum Profit_t - penalty \times (\sum Loss_t + 1/5 \times \sum Cost_t)}{c_t} \quad (8)$$

The *penalty* parameter in the reward function is also intended to modulate the significance of losses and transaction costs. It can be set according to the investor's risk aversion level.

3.5. General Assumptions

In many research, including [Kwan \(1999\)](#), [Zhang et al. \(2020\)](#) and [Zhao et al. \(2023\)](#), two core assumptions are typically made for such models:

- **Full Liquidity:** Transactions are assumed to be executed instantly upon decision.
- **Market Neutrality:** Transactions are assumed not to affect future asset prices.

Furthermore, although collateral is usually required for loans, we assume it is unnecessary in our context.

3.6. Benchmark Models

We employ return as the reward function to benchmark our model using the same agent and environment. also, the use of return-to-risk based reward functions is precluded due to lending opportunities in the environment.

Additionally, we select three benchmarks from traditional SPPO models, inspired by [Ngo et al. \(2023\)](#) and [Jin \(2023\)](#). SPPO models can be compared to our model by applying weight rebalancing in each state of the environment. Differences in risk calculation lead to distinct EFs, with optimal weights determined at the intersection of the EF and the investor's utility function or $w_t^{m+1} = 1$ if no intersection exists.

3.6.1. Return-based RL

Return (Equation 9) is a favored reward function in the RL-based portfolio management models due to its simplicity, adaptability to various environments, and alignment with the primary goal of maximizing the portfolio value (p).

$$Return = \frac{p_t}{p_{t-1}} - 1 \quad (9)$$

3.6.2. Mean-Variance (MV)

The MV model ([Markowitz \(1952\)](#)) is a computational framework used to select portfolio weights based on assessing risk to the expected return. In this model, portfolio risk (variance) is calculated using Equation 10.

$$\sigma_P = \sum_i w_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} w_i w_j \sigma_i \sigma_j \rho_{ij} \quad (10)$$

where w_i and σ_i are the weight and standard deviation of the i th asset, respectively, and ρ_{ij} is the correlation coefficient between assets i and j . Additionally, the EF in this model is derived using quadratic programming.

3.6.3. Mean Absolute Deviation (MAD)

MAD, a risk measure introduced by Konno and Yamazaki (1991) for portfolio management, quantifies the mean absolute deviation of individual asset returns from the portfolio's average return. It captures return dispersion and provides a robust risk measure less affected by extreme values. MAD is calculated using Equation 11, and the EF is determined using linear programming.

$$MAD_P = \frac{1}{n} \sum_{i=1}^n |r_i - E(r_P)| \quad (11)$$

where n is the number of assets in the portfolio, r_i and $E(r_P)$ are the return of the i th asset and the mean return of the portfolio, respectively.

3.6.4. Conditional Value at Risk (CVaR)

CVaR is a risk measure that assesses potential losses beyond a specific confidence level. It quantifies the expected loss exceeding the confidence level and highlights extreme loss severity, focusing on tail risk. CVaR aids in making informed decisions to mitigate potential downside risks in the portfolio management model introduced by Rockafellar et al. (2000). This model employs quadratic programming to derive the EF, with CVaR being determined through Equations 12 and 13.

$$CVaR = E(r_P | r_P \geq VaR^\alpha) \quad (12)$$

$$VaR^\alpha = F^{-1}(1 - \alpha) \quad (13)$$

where VaR^α represents the maximum expected loss with confidence level α , assuming portfolio returns (r_P) follow a normal distribution.

4. Empirical Studies

4.1. Setup Environment and Agent

In the model's implementation phase, we chose a diversified 12-crypto asset portfolio from the Binance perpetual futures market based on their liquidity and data availability. USDT serves as the portfolio's base asset, capable of granting and receiving loans. Our analysis covers two distinct periods: a high-volatility period from May 1, 2021, to September 1, 2022 (Port A), and a low-volatility period from June 1, 2022, to October 1, 2023 (Port B). Each period lasts 16 months, divided into 12 months (2178 steps) for training and 4 months (732 steps) for testing, featuring 4-hour rebalancing intervals. Additionally, hourly historical price data for cryptocurrencies is sourced from www.cryptodatadownload.com.

Figures 3 and 4 illustrate the investment value of one dollar across each currency within Portfolios A and B, respectively, during training and testing, and Table 1 provides statistical data on the 4-hour return distributions of the currencies.

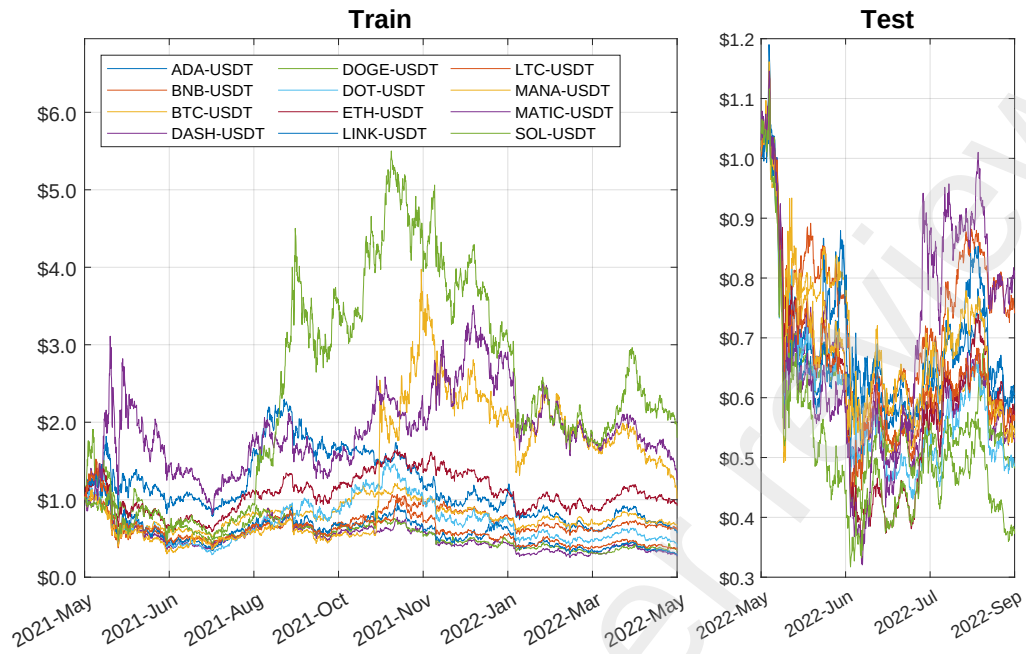


Figure 3: Investment value of one dollar across each currency in Port A during training and testing.

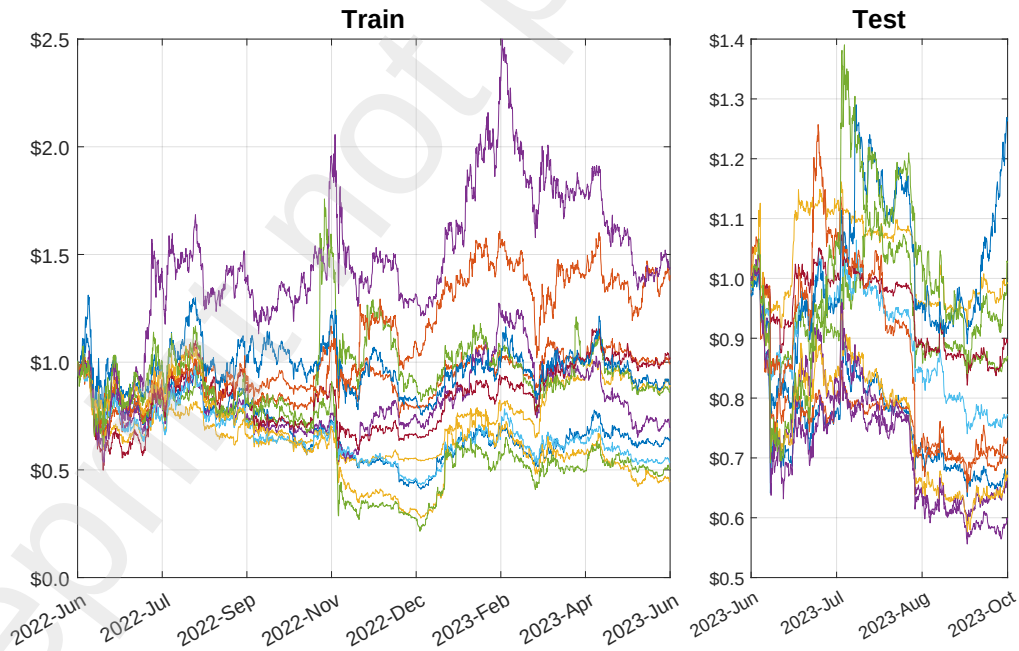


Figure 4: Investment value of one dollar across each currency in Port B during training and testing.

Table 1: Statistics for 4-hour return distributions of the currencies in Portfolios A and B.

Currencies INFO	Port A									
	Mean(%)	Std(%)	Min(%)	Max(%)	Total(%)	Mean(%)	Std(%)	Min(%)	Max(%)	Total(%)
	Train					Test				
ADA-USDT	0.00	2.37	-11.09	12.70	-43.35	-0.04	2.37	-14.88	15.45	-41.77
BNB-USDT	0.00	2.09	-13.82	10.59	-39.40	-0.02	1.82	-6.19	14.34	-27.18
BTC-USDT	-0.01	1.59	-10.40	8.22	-34.75	-0.07	1.54	-6.61	7.59	-47.12
DASH-USDT	-0.03	2.56	-13.40	16.46	-73.30	-0.06	2.16	-10.47	11.68	-48.77
DOGE-USDT	-0.01	2.98	-15.77	39.31	-66.51	-0.08	2.09	-11.56	12.85	-52.54
DOT-USDT	0.00	2.79	-24.95	21.46	-61.36	-0.07	2.37	-10.83	13.06	-52.64
ETH-USDT	0.02	2.05	-10.68	9.67	-10.03	-0.05	2.15	-7.57	10.42	-44.04
LINK-USDT	-0.02	2.74	-13.69	13.26	-72.45	-0.04	2.41	-9.10	9.38	-40.27
LTC-USDT	-0.02	2.28	-14.78	12.27	-64.87	-0.06	2.07	-8.49	10.43	-44.01
MANA-USDT	0.06	3.45	-17.03	41.11	0.94	-0.04	3.09	-10.85	42.23	-46.33
MATIC-USDT	0.06	3.21	-15.37	21.31	25.92	0.01	2.87	-12.34	13.82	-20.92
SOL-USDT	0.07	3.01	-16.90	17.12	77.00	-0.09	2.82	-13.92	13.89	-64.18
Port B										
ADA-USDT	-0.01	1.61	-7.52	10.80	-36.39	-0.04	1.40	-13.17	11.76	-30.10
BNB-USDT	0.01	1.27	-5.74	11.74	0.02	-0.04	0.96	-8.39	5.00	-29.42
BTC-USDT	0.00	1.19	-7.52	8.67	-11.02	0.00	0.72	-4.59	5.58	0.67
DASH-USDT	0.00	1.71	-12.30	13.08	-28.16	-0.04	1.47	-18.53	6.43	-32.96
DOGE-USDT	0.01	2.06	-12.73	29.36	-12.99	-0.01	1.26	-11.33	8.30	-13.06
DOT-USDT	-0.02	1.60	-8.72	10.27	-45.87	-0.03	1.09	-9.91	5.68	-21.35
ETH-USDT	0.01	1.59	-10.19	11.35	3.00	-0.01	0.75	-5.21	4.54	-9.76
LINK-USDT	0.01	1.82	-13.54	11.89	-7.60	0.04	1.41	-11.58	12.14	28.22
LTC-USDT	0.03	1.75	-14.58	15.76	41.39	-0.03	1.43	-13.04	13.19	-27.63
MANA-USDT	-0.02	1.95	-10.47	21.59	-54.23	-0.04	1.46	-11.02	7.36	-33.17
MATIC-USDT	0.04	2.09	-10.58	20.14	46.16	-0.06	1.46	-13.86	7.12	-39.61
SOL-USDT	0.00	2.46	-26.03	24.70	-48.87	0.02	1.75	-11.59	11.05	3.19

Following the methodology in section 3, the environment’s hyperparameters are detailed in Table 2. We utilized real information from the Binance exchange and identified optimal parameter values for achieving satisfactory results. The environment is designed to rebalance portfolio weights every 4 hours, enhancing the profit-to-cost ratio. Shorter intervals resulted in increased trading costs without generating considerable additional profit, which is undesirable. Additionally, the analysis of the last 48 hours (i.e. $N = 49$) of historical data enhanced the agent’s market condition understanding. Utilizing data beyond 48 hours risked overemphasis on past market behavior. Furthermore, the penalty parameter in the reward function, aimed at controlling downside risk, is set to 25. This indicates the agent’s sensitivity to loss is 25 times greater than profit.

Table 2: Environment’s hyperparameters.

Hyperparameters	Value
Initial investment	1000\$
Historical data length	48 hours
Rebalancing frequency	4 hours
Penalty parameter	25
Annual interest rate for borrowing USDT	5%
Annual interest rate for lending USDT	3%
Transaction fee	0.05%

The SAC agent’s optimal hyperparameters are also outlined in Table 3, fine-tuned to accelerate response convergence during training.

Table 3: The SAC agent’s hyperparameters.

Hyperparameters	Value
Optimizer	Adam
Actor Learning Rate	2e-4
Critic Learning Rate	6e-4
Discount Factor	0.99
Entropy Weight	0.08
Entropy Learn Rate	6e-4
L2RegularizationFactor	1e-8
Mini Batch Size	16
Sample Time	1
Steps to Look Ahead	1

4.2. Testing Results

We tested our trained agent against return-based RL and SPPO benchmarks with a standard risk aversion parameter (i.e., $riskaversion = 4$) as introduced in subsection 3.6. Figures 5 and 6 present the 4-month testing results for Portfolios A

and B, respectively. Meanwhile, Table 4 provides a performance comparison based on the criteria outlined in the Wood et al. (2021)'s paper.

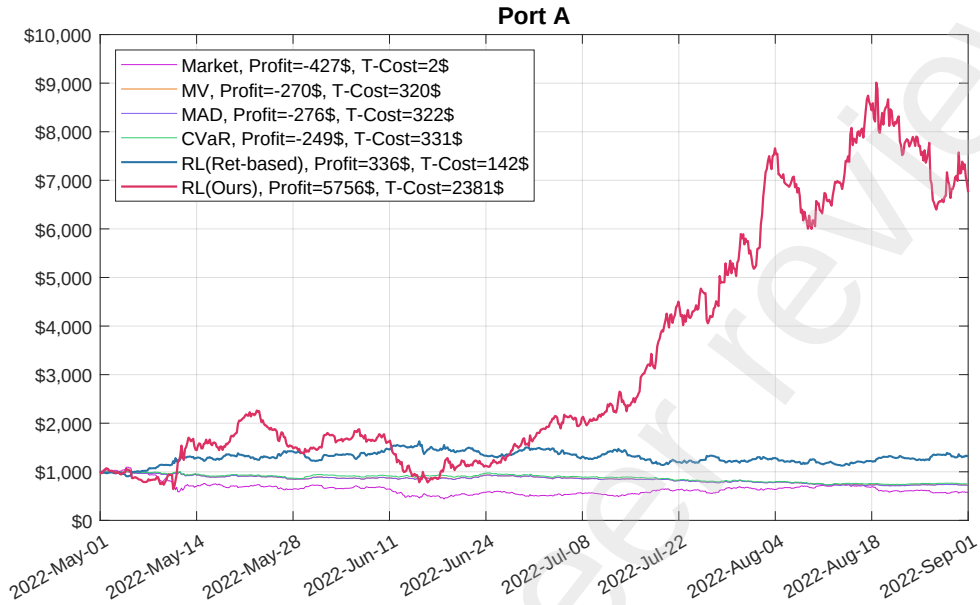


Figure 5: Testing results for Port A, Market represents an equally weighted strategy.

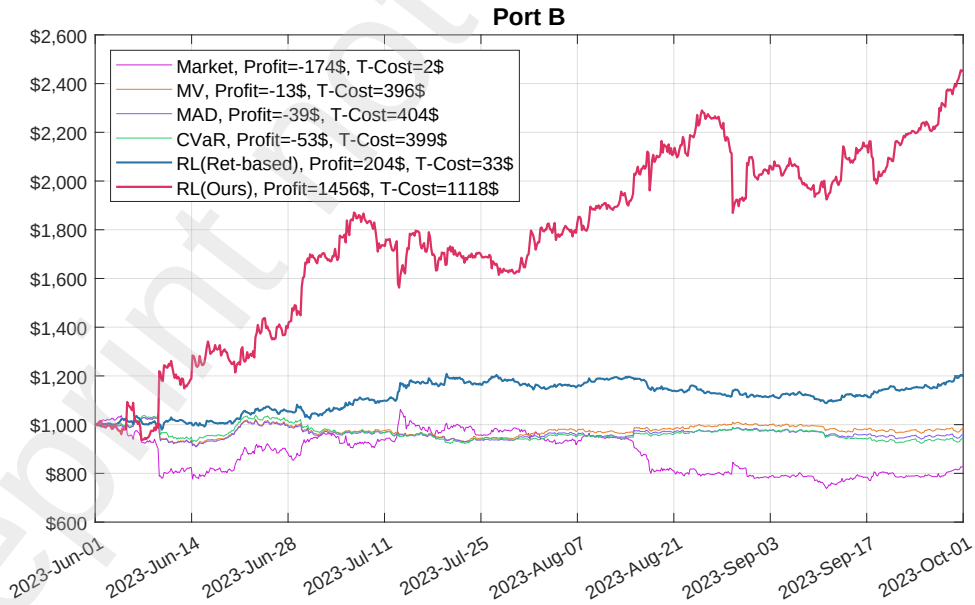


Figure 6: Testing results for Port B, Market represents an equally weighted strategy.

Table 4: Comparison of the testing results for Portfolios A and B.

Evaluation Metrics	Total Return(%)	Win Rate(%)	Average Return(%)	Standard Deviation(%)	Sharpe Ratio	Downside Deviation(%)	Sortino Ratio	Max Drawdown(%)	Calmar Ratio
<i>Port A</i>									
MV	-26.978	49.379	-0.040	0.679	-1.487	0.513	-1.970	28.547	-0.961
MAD	-27.604	49.481	-0.042	0.681	-1.518	0.518	-1.995	29.335	-0.957
CVaR(95%)	-24.890	49.413	-0.036	0.697	-1.334	0.519	-1.792	27.676	-0.912
RL(Ret-based)	33.573	49.282	0.053	1.671	1.021	1.120	1.523	30.859	1.502
R(Ours)	575.585	52.840	0.347	4.181	10.468	3.031	14.442	65.519	18.149
<i>Port B</i>									
MV	-1.324	49.078	-0.001	0.461	-0.178	0.456	-0.180	11.408	-0.195
MAD	-3.911	48.896	-0.004	0.458	-0.390	0.465	-0.385	11.739	-0.412
CVaR(95%)	-5.328	49.192	-0.006	0.447	-0.522	0.422	-0.553	11.535	-0.547
RL(Ret-based)	20.384	49.453	0.026	0.541	1.368	0.309	2.388	9.961	2.011
RL(Ours)	145.627	51.730	0.139	1.815	3.557	1.187	5.438	18.370	9.507

testing results reveal our model outperformed benchmark models, achieving total returns of 575.5% and 145.6% in Portfolios A and B, respectively, compared to non-converged return-based RL with a low positive return and SPPO models that only succeed in reducing market losses (equally weighted strategy). The non-convergence suggests that the return reward function is ineffective in guiding the agent to optimal solutions.

Despite facing higher risks due to increased portfolio value, our model demonstrated superior return-to-risk ratios, particularly in the Sortino and Kalmar ratios compared to the Sharpe ratio. This underscores its proficiency in managing downside risk. Additionally, the model excelled in the high-volatility period (Port A), illustrating its ability to capitalize on market fluctuations for higher profit while efficiently managing risk.

Table 4 also notes a nearly identical win rate (accuracy in predicting market direction) across all models. Thus, the marked difference in total returns can be attributed to the unique weight optimizations utilized by each model. RL models dynamically rebalance portfolio weights based on environmental insights and future event projections, prioritizing more profitable assets. In this context, our model exhibits greater accuracy than return-based RL due to convergence during the learning process. Conversely, SPPO benchmarks, which base weight rebalancing on historical data, often fail to adapt to market volatility, particularly in highly volatile markets like cryptocurrency.

Figures 7 and 8 complement this discussion by presenting the normal-curve-fitted distribution of the product of weights in the corresponding asset returns for our model alongside the top-performing benchmarks in Portfolios A and B during testing.

These distributions reveal a right-skewed profit distribution for our models, with kurtosis closer to zero for losses than profits. Conversely, the return-based RL, MV and CVaR models' distributions display kurtosis closer to zero for profits than for losses and left-skewed loss distribution, notably in the MV model.

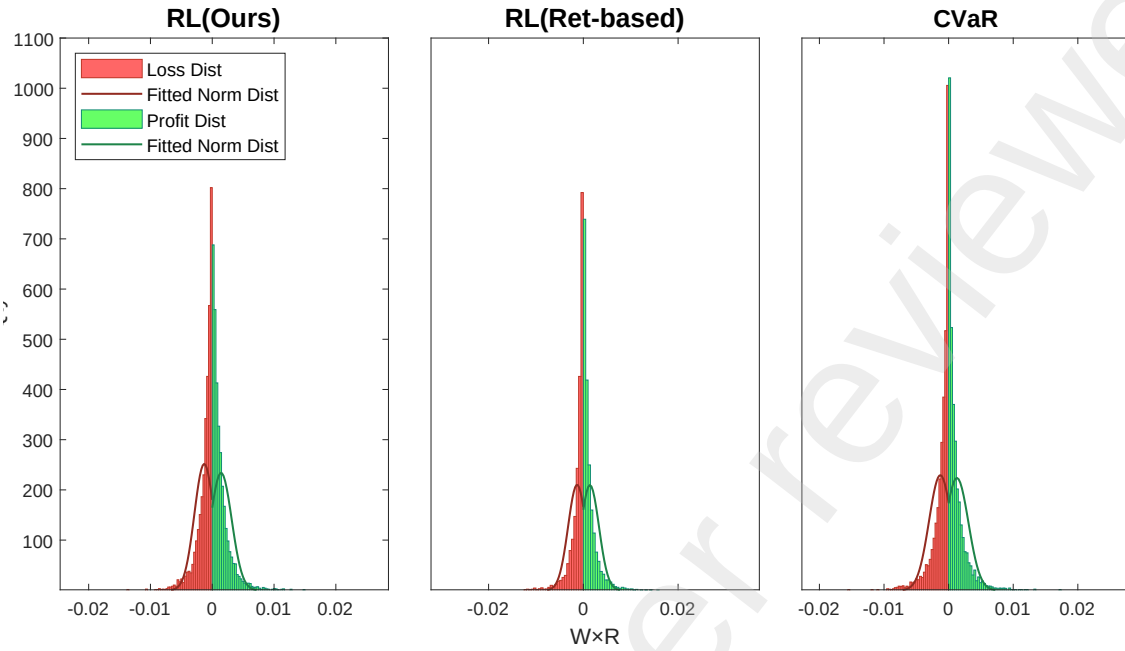


Figure 7: Distribution of $weight \times return$ and fitted normal distribution in Port A during testing.

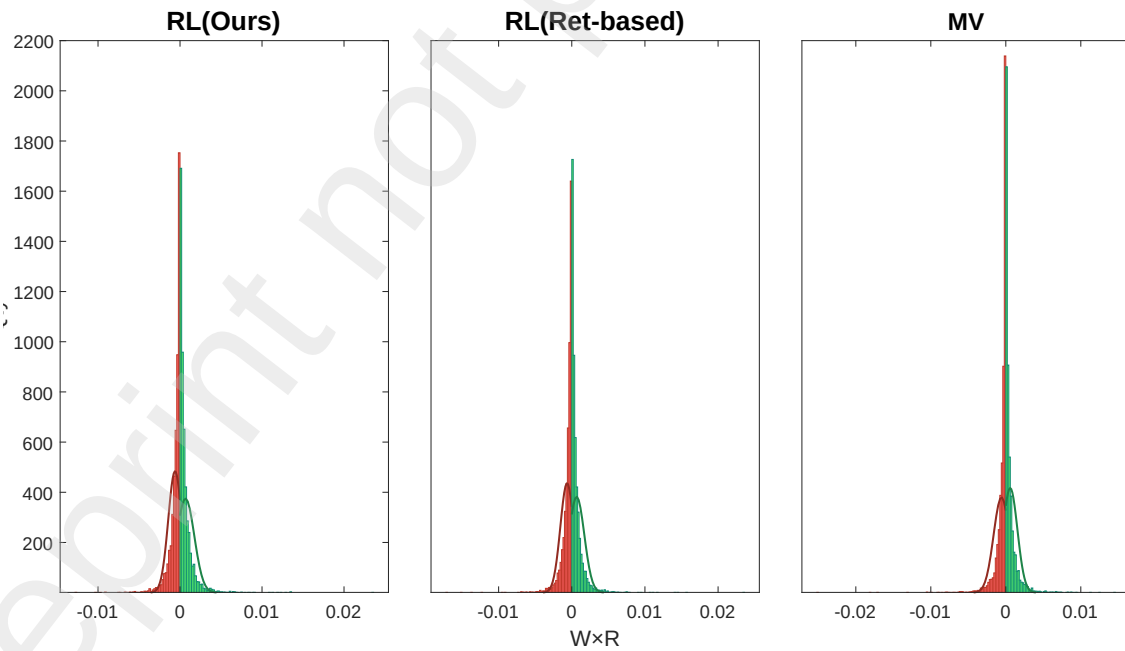


Figure 8: Distribution of $weight \times return$ and fitted normal distribution in Port B during testing.

Therefore, despite having comparable win rates, benchmark models fail to achieve high profits and manage risk adequately.

5. Conclusion

In summary, this study introduces an RL-based portfolio management model tailored for high-risk settings, overcoming the limitations of traditional RL frameworks by leveraging two-sided transactions and lending. We developed a novel environmental formulation and a PnL-based reward function to improve the agent's convergence in training by enhancing its understanding of the additional risks presented by new features in the environment.

Testing results in the challenging cryptocurrency market indicate that our model surpasses benchmarks, including return-based RL, MV, MAD, and CVaR models, effectively managing a diversified 12-crypto asset portfolio across different market volatility. Additionally, our model demonstrated superior return-to-risk ratios, particularly in the Sortino and Kalmar ratios compared to the Sharpe ratio. This underscores its proficiency in managing downside risk. Furthermore, the model excelled during the high-volatility period, illustrating its ability to capitalize on market fluctuations for higher profit while efficiently managing risk.

This study highlights two main findings: First, our model adeptly incorporates two-sided transactions and lending in weight optimization to exploit all possible market opportunities, making it suitable for investors seeking profits under various market conditions. Second, by moving beyond the preset assumptions of SPPO models, our approach accommodates future asset price uncertainty, allowing for dynamic portfolio rebalancing and capturing the nonlinear effects of multiple asset risks to improve profitability.

Thus, the presented model offers a promising, state-of-the-art framework for investors aiming to achieve profit in all market conditions and secure higher return-to-risk ratios.

References

- A. M. Aboussalah, Z. Xu, and C.-G. Lee. What is the value of the cross-sectional approach to deep reinforcement learning? *Quantitative Finance*, 22(6):1091–1111, 2022. [4](#), [5](#)
- D. Barro, G. Consigli, and V. Varun. A stochastic programming model for dynamic portfolio management with financial derivatives. *Journal of Banking & Finance*, 140:106445, 2022. [4](#)
- R. Barua and A. K. Sharma. Dynamic black litterman portfolios with views derived via cnn-bilstm predictions. *Finance Research Letters*, 49:103111, 2022. [2](#)
- R. Barua and A. K. Sharma. Using fear, greed and machine learning for optimizing global portfolios: A black-litterman approach. *Finance Research Letters*, 58:104515, 2023. [2](#)
- C. Betancourt and W.-H. Chen. Deep reinforcement learning for portfolio management of markets with a dynamic number of assets. *Expert Systems with Applications*, 164:114002, 2021. [5](#)
- N. Borri. Conditional tail-risk in cryptocurrency markets. *Journal of Empirical Finance*, 50:1–19, 2019. [3](#)
- A. H. Chen, F. C. Jen, and S. Zions. The optimal portfolio revision policy. *Journal of business*, pages 51–61, 1971. [4](#)
- C.-H. Cheng and L.-Y. Wei. A novel time-series model based on empirical mode decomposition for forecasting taiaex. *Economic Modelling*, 36:136–141, 2014. [5](#)
- A. Coache and S. Jaimungal. Reinforcement learning with dynamic convex risk measures. *Mathematical Finance*, 2023. [4](#)
- T. Cui, S. Ding, H. Jin, and Y. Zhang. Portfolio constructions in cryptocurrency market: A cvar-based deep reinforcement learning approach. *Economic Modelling*, 119:106078, 2023. [2](#), [5](#), [6](#)
- X. Cui, D. Li, and X. Li. Mean-variance policy for discrete-time cone-constrained markets: Time consistency in efficiency and the minimum-variance signed supermartingale measure. *Mathematical Finance*, 27(2):471–504, 2017. [4](#)
- S. R. Das, D. Ostrov, A. Radhakrishnan, and D. Srivastav. Dynamic optimization for multi-goals wealth management. *Journal of Banking & Finance*, 140:106192, 2022. [4](#)

- E. F. Fama. Multiperiod consumption-investment decisions. *Stochastic optimization models in finance*, pages 389–400, 1975. [4](#)
- F. Fang, C. Ventre, M. Basios, L. Kanthan, D. Martinez-Rego, F. Wu, and L. Li. Cryptocurrency trading: a comprehensive survey. *Financial Innovation*, 8(1): 1–59, 2022. [3](#)
- G. Faria and F. Verona. Forecasting stock market returns by summing the frequency-decomposed parts. *Journal of Empirical Finance*, 45:228–242, 2018. [5](#)
- L. V. Fulton and N. D. Bastian. Multiperiod stochastic programming portfolio optimization for diversified funds. *International Journal of Finance & Economics*, 24(1):313–327, 2019. [4](#)
- A. A. Gaivoronski and F. Stella. On-line portfolio selection using stochastic programming. *Journal of Economic Dynamics and Control*, 27(6):1013–1043, 2003. [4](#)
- X. Gao and L. Chan. An algorithm for trading and portfolio management using q-learning and sharpe ratio maximization. In *Proceedings of the international conference on neural information processing*, pages 832–837. Citeseer, 2000. [4](#)
- S. Goutte, H.-V. Le, F. Liu, and H.-J. Von Mettenheim. Deep learning and technical analysis in cryptocurrency market. *Finance Research Letters*, 54:103809, 2023. [5](#)
- N. Gradojevic and R. Gençay. Fuzzy logic, trading uncertainty and technical trading. *Journal of Banking & Finance*, 37(2):578–586, 2013. [5](#)
- T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta, P. Abbeel, et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018. [2](#)
- I. Halperin. The qlbs q-learner goes nuclear: fitted q iteration, inverse rl, and option portfolios. *Quantitative Finance*, 19(9):1543–1553, 2019. [4](#), [5](#)
- B. Hambly, R. Xu, and H. Yang. Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503, 2023. [4](#), [5](#)
- T. Jaisson. Deep differentiable reinforcement learning and optimal trading. *Quantitative Finance*, 22(8):1429–1443, 2022. [4](#), [5](#)
- J. Jang and N. Seong. Deep reinforcement learning for stock portfolio optimization by connecting with modern portfolio theory. *Expert Systems with Applications*, 218:119556, 2023. [5](#)

- B. Jin. A mean-var based deep reinforcement learning framework for practical algorithmic trading. *IEEE Access*, 11:28920–28933, 2023. 4, 10
- P. Jorion. Value at risk: the new benchmark for controlling market risk. (*No Title*), 1997. 3
- J. L. Kelly. A new interpretation of information rate. *the bell system technical journal*, 35(4):917–926, 1956. 3
- H. Konno and H. Yamazaki. Mean-absolute deviation portfolio optimization model and its applications to tokyo stock market. *Management science*, 37(5):519–531, 1991. 3, 11
- H. Kraft and M. Steffensen. A dynamic programming approach to constrained portfolios. *European Journal of Operational Research*, 229(2):453–461, 2013. 4
- C. C. Kwan. A note on market-neutral portfolio selection. *Journal of Banking & Finance*, 23(5):773–800, 1999. 10
- J. Lee, H. Koh, and H. J. Choe. Learning to trade in financial time series using high-frequency through wavelet transformation and deep reinforcement learning. *Applied Intelligence*, 51:6202–6223, 2021. 5
- D. Li and W.-L. Ng. Optimal dynamic portfolio selection: Multiperiod mean-variance formulation. *Mathematical finance*, 10(3):387–406, 2000. 4
- J. Li, Y. Zhang, X. Yang, and L. Chen. Online portfolio management via deep reinforcement learning with high-frequency data. *Information Processing & Management*, 60(3):103247, 2023a. 2, 4, 5
- S. Li, X. Huang, Z. Cheng, W. Zou, and Y. Yi. Ae-acg: A novel deep learning-based method for stock price movement prediction. *Finance Research Letters*, 58: 104304, 2023b. 3
- W. Lin and A. Taamouti. Portfolio selection under non-gaussianity and systemic risk: A machine learning based forecasting approach. *International Journal of Forecasting*, 2023. 2
- J. Liu, L. Lu, X. Zong, and B. Xie. Nonlinear relationships in soybean commodities pairs trading-test by deep reinforcement learning. *Finance Research Letters*, 58: 104477, 2023a. 5
- Q. Liu, Z. Tao, Y. Tse, and C. Wang. Stock market prediction with deep learning: The case of china. *Finance Research Letters*, 46:102209, 2022. 5

- Y. Liu, Z. Li, R. Nekhili, and J. Sultan. Forecasting cryptocurrency returns with machine learning. *Research in International Business and Finance*, 64:101905, 2023b. [2](#)
- S. Luo and X. Zeng. An optimal investment model with markov-driven volatilities. *Quantitative Finance*, 14(9):1651–1661, 2014. [4](#)
- C. Ma and S. Yan. Deep learning in the chinese stock market: The role of technical indicators. *Finance Research Letters*, 49:103025, 2022. [5](#)
- H. M. Markovitz. *Portfolio selection: Efficient diversification of investments*. John Wiley, 1959. [3](#)
- H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. [3](#), [10](#)
- R. C. Merton. Lifetime portfolio selection under uncertainty: The continuous-time case. *The review of Economics and Statistics*, pages 247–257, 1969. [4](#)
- R. O. Michaud. The markowitz optimization enigma: Is ‘optimized’ optimal? *Financial analysts journal*, 45(1):31–42, 1989. [3](#)
- J. Moody and M. Saffell. Learning to trade via direct reinforcement. *IEEE transactions on neural Networks*, 12(4):875–889, 2001. [2](#), [4](#)
- J. Mossin. Optimal multiperiod portfolio policies. *The Journal of Business*, 41(2): 215–229, 1968. [4](#)
- O. Mounjid and C.-A. Lehalle. Improving reinforcement learning algorithms: towards optimal learning rate policies. *Mathematical Finance*, 2023. [4](#)
- J. M. Mulvey, Y. Sun, M. Wang, and J. Ye. Optimizing a portfolio of mean-reverting assets with transaction costs via a feedforward neural network. *Quantitative Finance*, 20(8):1239–1261, 2020. [2](#)
- R. Neuneier. Enhancing q-learning for optimal asset allocation. *Advances in neural information processing systems*, 10, 1997. [4](#)
- R. Neuneier and O. Mihatsch. Risk sensitive reinforcement learning. *Advances in neural information processing systems*, 11, 1998. [4](#)
- V. M. Ngo, H. H. Nguyen, and P. Van Nguyen. Does reinforcement learning outperform deep learning and traditional portfolio optimization models in frontier and developed financial markets? *Research in International Business and Finance*, 65:101936, 2023. [5](#), [10](#)

- S. Petchrompo, A. Wannakrairot, and A. K. Parlikad. Pruning pareto optimal solutions for multi-objective portfolio asset management. *European Journal of Operational Research*, 297(1):203–220, 2022. [3](#)
- Y.-S. Ren, C.-Q. Ma, X.-L. Kong, K. Baltas, and Q. Zureigat. Past, present, and future of the application of machine learning in cryptocurrency research. *Research in International Business and Finance*, 63:101799, 2022. [3](#)
- R. T. Rockafellar, S. Uryasev, et al. Optimization of conditional value-at-risk. *Journal of risk*, 2:21–42, 2000. [3](#), [11](#)
- D. Roman, K. Darby-Dowman, and G. Mitra. Mean-risk models using two risk measures: a multi-objective approach. *Quantitative Finance*, 7(4):443–458, 2007. [3](#)
- P. Samuelson. The fundamental approximation theorem of portfolio analysis in terms of means variances and higher moments. *Review of Economic Studies*, 25: 65–86, 1958. [3](#)
- K. V. Smith. A transition model for portfolio revision. *The Journal of Finance*, 22 (3):425–439, 1967. [4](#)
- T. Stein. Forecasting the equity premium with frequency-decomposed technical indicators. *International Journal of Forecasting*, 40(1):6–28, 2024. [5](#)
- R. S. Sutton and A. G. Barto. *Reinforcement learning: An introduction*. MIT press, 2018. [2](#)
- X. Tan and Y. Tao. Trend-based forecast of cryptocurrency returns. *Economic Modelling*, 124:106323, 2023. [5](#)
- P. Tang, C. Tang, and K. Wang. Stock movement prediction: A multi-input lstm approach. *Journal of Forecasting*, pages 1–13, 2024. [5](#)
- N. Topaloglou, H. Vladimirov, and S. A. Zenios. A dynamic stochastic programming model for international portfolio management. *European Journal of Operational Research*, 185(3):1501–1524, 2008. [4](#)
- C.-Y. Tsao. Portfolio selection based on the mean–var efficient frontier. *Quantitative Finance*, 10(8):931–945, 2010. [3](#)
- P. M. Van Staden, D.-M. Dang, and P. A. Forsyth. Time-consistent mean–variance portfolio optimization: A numerical impulse control approach. *Insurance: Mathematics and Economics*, 83:9–28, 2018. [4](#)

- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 3
- H. Wang and X. Y. Zhou. Continuous-time mean–variance portfolio selection: A reinforcement learning framework. *Mathematical Finance*, 30(4):1273–1308, 2020. 4
- J. Wang, G. Athanasopoulos, R. J. Hyndman, and S. Wang. Crude oil price forecasting based on internet concern using an extreme learning machine. *International Journal of Forecasting*, 34(4):665–677, 2018. 5
- X. Wang, Y. Ouyang, Y. Li, S. Liu, L. Teng, and B. Wang. Multi-objective portfolio selection considering expected and total utility. *Finance Research Letters*, 58: 104552, 2023. 3
- K. Wood, S. Roberts, and S. Zohren. Slow momentum with fast reversion: A trading strategy using deep learning and changepoint detection. *The Journal of Financial Data Science*, 2021. 15
- M.-E. Wu, J.-H. Syu, J. C.-W. Lin, and J.-M. Ho. Portfolio management system in equity market neutral using reinforcement learning. *Applied Intelligence*, pages 1–13, 2021. 4
- K. Xu and W. Wang. Limited information limits accuracy: Whether ensemble empirical mode decomposition improves crude oil spot price prediction? *International Review of Financial Analysis*, 87:102625, 2023. 5
- M. R. Young. A minimax portfolio selection rule with linear programming solution. *Management science*, 44(5):673–683, 1998. 3
- Z. Zhang, S. Zohren, and R. Stephen. Deep reinforcement learning for trading. *The Journal of Financial Data Science*, 2020. 10
- T. Zhao, X. Ma, X. Li, and C. Zhang. Asset correlation based deep reinforcement learning for the portfolio selection. *Expert Systems with Applications*, 221:119707, 2023. 4, 10
- X. Y. Zhou and D. Li. Continuous-time mean-variance portfolio selection: A stochastic lq framework. *Applied Mathematics and Optimization*, 42:19–33, 2000. 4